Jorge Condor

PhD Candidate
USI Lugano, Switzerland

- October 10, 1998
- Spanish
- jorge.condor@usi.ch
- +41 078 224 70 97
- https://arcanous98.github.io/

Languages

- Spanish
- English
- Italian
- French

Toolbox

- Programming Languages (C++, Python, MATLAB, CUDA, Slang)
- Machine learning frameworks (PyTorch)
- Physically-based path tracing and differentiable rendering (Mitsuba 3, OptiX)
- Neural Reconstruction and Rendering (3D Gaussian Splatting, NeRF)

Education

PhD

March 2022 – Present — **PhD Student, IDSIA-USI** — The Swiss AI Lab IDSIA-USI Lugano, Switzerland

PhD Candidate in Computer Graphics at IDSIA-USI under Prof. Piotr Didyk. My research bridges inverse rendering, neural representations and physically-based rendering, with applications in vision, real-time graphics and fabrication.

Master

September 2020 – February 2022 — **Master of Engineering in Robotics, Graphics and Computer Vision (English)** — Universidad de Zaragoza

Honors in Modelling and Simulation of Appearance (PBR course), where I developed a path tracer with support for heterogeneous volumetric multiple scattering and fur rendering. Honorable Mention in the Rendering Contest judged by Marcos Fajardo, Matt Chiang and Wojciech Jarosz. Honors in my Master Thesis on Neural Rendering of Complex Luminaires under Prof. Adrián Jarabo (Graphics & Imaging Lab), published at EGSR 2022.

Bachelor

September 2016 – July 2020 — **Bachelor in Electronics and Automatic Control Engineering (Spanish)** — Universidad de Zaragoza

Top 7% of my class. Bachelor Thesis on Deep Learning for localization and classification of microparticles from digital holograms (Optical Laser Technology Group, I3A). Erasmus year (2019–2020) at **Aalto University, Finland**, taking Master-level courses on AI, electronics design and robotics. Honors in numerous courses, covering digital electronics, thermodynamics and chemistry.

Working Experience

From Aug. 2026 — **Research Scientist Intern (incoming)** — Meta Reality Labs, Zurich, Switzerland

Collaborating with the Hyperscape team to work on neural reconstruction and rendering, under the supervision of Adrián Jarabo.

June 2025 – May 2026 — **Research Scientist Intern** — NVIDIA, Zurich, Switzerland

Working with the Spatial Intelligence Lab under Zan Gojcic on Neural Reconstruction and Rendering.

May 2024 – August 2024 — **External Research Collaborator @ Meta** — Nonstop Consulting (Remote)

Part-time role to continue my work alongside the Monaco rendering team at Meta, working on physically based rendering.

September 2023 – January 2024 — **Research Scientist Intern** — Meta Reality Labs, Zurich, Switzerland

Worked with the Monaco rendering team on physically-based rendering involving primitive path tracing, Gaussian splatting and differentiable rendering, under the supervision of Christophe Hery and Adrián Jarabo

September 2022 – Present — **Teaching Assistant, Computer Graphics and Image and Video Processing Courses** — USI, Switzerland

Teaching assistant for the bachelor-level Computer Graphics course and the master-level Image and Video Processing course in USI. My duties include design of practical assignments, preparation of practical exercise lessons, grading of assignments and exams and attending students questions over theory or coding.

February 2021 – February 2022 — **Research Intern, Graphics and Imaging Lab** — Universidad de Zaragoza, Spain

Collaborated on projects related to image-based material appearance editing and neural path tracing. Works published at Computer Graphics Forum (2021) and Eurographics Symposium on Rendering (2022).

July 2017 – September 2019 — **Mathematics, Physics and Chemistry Tutor** — Zaragoza, Spain

Mathematics, Physics and Chemistry tutor for baccalaureate (university entry exams preparation) students

Publications

Apr 2026 ★ **Neural Harmonic Textures for High-Quality Primitive Based Neural Reconstruction**
Jorge Condor*, Nicolas Moenne-Loccoz, Merlin Nimier-David, Piotr Didyk, Zan Gojcic, Qi Wu*
arXiv preprint

Feb 2026 ★ **Gabor Fields: Orientation-Selective Level-of-Detail for Volume Rendering**
Jorge Condor**, Nicolai Hermann*, Mehmet Ata Yurtsever, Piotr Didyk* *(*joint first authors)*
ACM Transactions on Graphics (SIGGRAPH'26)

Aug 2025 **Human Vision Constrained Super-Resolution**
Volodymyr Karpenko, Taimoor Tariq, ***Jorge Condor****, Piotr Didyk*
Workshop on Human-Inspired Computer Vision (ICCV Workshops'25)

Jul 2025 **Puzzle Similarity: A Perceptually-guided No-Reference Metric for Artifact Detection in 3D Scene Reconstructions**
Nicolai Hermann, ***Jorge Condor****, Piotr Didyk*
International Conference on Computer Vision (ICCV'25)

Jun 2025 **Temporal Brightness Management for Immersive Content**
Luca Surace, ***Jorge Condor****, Piotr Didyk*
Eurographics Symposium on Rendering (EGSR'25)

Dec 2024 ★ **Don't Splat your Gaussians: Volumetric Ray-Traced Primitives for Modeling and Rendering Scattering and Emissive Media**
Jorge Condor*, Sébastien Speierer, Lukas Bode, Aljaž Božič, Simon Green, Piotr Didyk, Adrián Jarabo*
ACM Transactions on Graphics, 2024 (Presented at SIGGRAPH'25)

Jul 2023 ★ **Gloss-aware Color Correction for 3D Printing**
Jorge Condor*, Michal Piovarci, Bernd Bickel, Piotr Didyk*
Proceedings of SIGGRAPH (SIGGRAPH'23)

Jul 2022 **A Learned Radiance-Field Representation for Complex Luminaires**
Jorge Condor*, Adrián Jarabo*
Eurographics Symposium on Rendering (EGSR'22)

Jan 2022 **A Generative Framework for Image-based Editing of Material Appearance using Perceptual Attributes**
Johanna Delanoy, Manuel Lagunas, ***Jorge Condor****, Diego Gutiérrez, Belén Masiá*
Computer Graphics Forum (Presented at EUROGRAPHICS'22)

Oct 2021 **Normal Map Estimation in the Wild**
Jorge Condor*, Manuel Lagunas, Johanna Delanoy, Belén Masiá, Diego Gutiérrez*
Poster, X Jornada Jóvenes Investigadores del I3A

Representative works marked with ★.

Supervision

Master Theses

Nov 2024 – Present **Kacper Kramarz-Fernández** MSc HPC (EU4HPC), USI
Efficient Physically-Based Rendering of Kernel-based Volumes.

Feb – Sept 2024 **Nicolai Hermann** MSc Artificial Intelligence, USI
Perceptually-Driven Neural Inpainting for Seamless 3D Reconstructions. *Best Master Thesis Award @ USI 2025.*

Feb 2023 – Sept 2024 **Michele Chersich** MSc Computational Science, USI
Reinforcement Learning for Importance Sampling of Neural Radiance Fields.

Research Interns

Sept 2025 – Present **Ewa Miazga** USI, Switzerland
Working on appearance modelling for neural representations. Work under submission.

Jul – Sept 2023 **Xiana Carrera Alonso** USI, Switzerland
Now a PhD student at Columbia University under Silvia Sellán.

Jul – Sept 2023 **Arnaud Fauconnet** USI, Switzerland
Now at Lifeware, Switzerland.

Dissemination

2026 **Invited Talks** Max Planck Institute für Informatik, Saarbrücken, Germany
On physically-based rendering and neural reconstruction.

2022 – Ongoing **Conference Talks** SIGGRAPH'23, SIGGRAPH'25, EGSR'22
Presented first-author works at flagship Computer Graphics venues (SIGGRAPH, EGSR), as well as poster sessions at ICCV, SIGGRAPH and the International Computer Vision Summer School.

2022 – 2025 **Highschool Maturity Projects (Mentor)** Liceo Cantonale 2, Lugano, Switzerland
Mentored highschool students through their *Lavoro di Maturità*: design of an IoT plant-health monitoring device based on LoRa and ESP32.

Diplomas & Scholarships

Scholarships

2021 **Beca TFM+Practicas** Zaragoza, Spain
Competitive scholarship awarding 600€/month to develop your master thesis within a research group

2020 **Strategic Master Scholarship** Zaragoza, Spain
Competitive scholarship awarding 4700€ to cover master studies

2019 **Santander Erasmus Scholarship** Zaragoza, Spain
Competitive scholarship awarding 500€ to cover Erasmus expenses

2014 **Aragonese Government Scholarship for a Linguistic Immersion in the English Language** Ontario, Canada
1-month stay in Ontario, Canada, studying in the F.E. Madill School. Granted for excellent results in high school studies

Diplomas

2017 **Driving Licence** Zaragoza, Spain
B Licence

2015 **Cambridge English Level 2 Certificate in ESOL International (Advanced C1)**
Overall score 199 (highest grading 202 in Speaking)